DRACO: Fine-Grained Credit Assignment with Dynamic Rubrics for Long-Horizon Agent Training

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HarnessRSI 66 ★★

recorded: retained

Paper

Code

1 What changes, what is kept

Retained artifact: LoRA policy weights
Not a prompt / tool / harness-code patch

1 Sample a rollout group
90 AppWorld training tasks; six rollouts per task; no ground-truth reward

2 Generate → merge → filter rubrics
Frozen GPT-5.4 uses instructions + traces
Keep criterion if any group member fails

3 Judge trajectories, then cite steps
Pass / fail / not applicable; separate attribution call may override verdicts

4 Route group-relative advantage
Winner: reinforce judge-approved steps
Loser: suppress judge-disapproved steps

5 Update the LoRA adapter via GRPO
Credit weights the gradient, not selection
No verifier-based update acceptance gate

Judge-based credit ≠ causal ground truth
Step totals length-normalized; sign preserved

2 Headline: held-out success (%)



Qwen3.6-27B · Table 2 p1. Mean of final 3 checkpoints (90/95/100), each evaluated 3 times; not training seeds. TN / TC / Banking: 168 / 417 / 97 tasks.

4 Where gains are absent or thin

- **TC SGC:** full 40.7 < static/no-credit 42.4. TC TGC p3: 43.9 < 45.6. Not every metric wins.
- **Banking:** 20.4 ± 1.0 vs dynamic/no-credit 19.7 ± 0.5; gap comparable to checkpoint SD. Self-judge highest: 21.1 ± 1.4.
- **Qwen2.5:** transfer 3.4 ± 0.6 → 4.0 ± 1.0. TN TGC 62.9 < SALT 66.2 (outcome-aware; unmatched budget).
- **TN hard SGC:** 51.9 < dynamic/no-credit 55.0. Extra gains strongest on TN, not uniformly OOD.

3 Controls, costs and judge validity

Crossed ablation · TN p1 (Table 2)

Rubric / credit	TGC	SGC
Static / none	81.1	59.9
Static / step credit	81.9	60.7
Dynamic / none	82.1	64.9
Dynamic / step credit	**85.3**	**70.6**

Static+credit gets step attribution; outcome GRPO does not. No outcome+credit control.

What signal does the judge use?
Instructions + traces, not gold outcomes. Causal credit not human-validated; judge bias possible. AppWorld: unit tests. Banking: GPT-5.4 simulator/judge.

Uncertainty & selection
TN TGC 85.3 ± 1.5 vs outcome 80.0 ± 1.9. SD: checkpoints, not training seeds. Average 55.7 vs 53.3: no SD/CI. Precommitted defaults; selection split/tuning budgets unreported. No test-best checkpoint selection.

Judge cost · 100 training steps
Full $1607 vs static/no-credit $570. Task $63‡; rollout $527‡; merge $318‡; attribution $308‡ (Fig. 8). Excludes GPU. Self-judge $316: thinking/k=3 per Fig. 8; §4.3 wording ambiguous. No weaker-variant cost identified.

Inference cost is separate: TN $10.77 → $8.27. Appendix B.3 "baseline" shorthand vs E.7: equal step totals ≠ uniform token credit.

Takeaway

Dynamic rubrics + step credit improve held-out AppWorld, especially TN scenario completion. Better task outcomes do not validate causal attribution. Transfer is less uniform, and equal training steps do not match judge cost.

Our read · HarnessRSI

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Reports: tests, transfer, ablations, cost, checkpoint SD.
Missing: training seeds, human validity, cost match, tested leakage audit.
Scope: LoRA weights, not harness-only RSI.